

Modeling Cognitive Load as an Instructional Partitioning Problem

A theory-first proposal on segmentation, coherence, and working-memory constraints

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EDPS 506 | Spring 2026

Known

The literature already gives us a strong chain of claims.

1 · **Capacity**

Working memory is a limited system for temporary storage and manipulation.

Baddeley, 1992

2 · **Load**

Learning depends on how intrinsic and extraneous demands consume that limited capacity.

Sweller, 1988; Sweller, 2010

3 · **Structure**

Task complexity depends on element interactivity and prior knowledge, not just item count.

Sweller et al., 2019; Chen et al., 2023

4 · **Optimization**

Capacity can be modeled as graded, allocable, and costly.

Ma et al., 2014; van den Berg & Ma, 2018

Gap

The missing piece is not whether load matters. It is how instructional structure becomes a prediction.

No explicit content structure

Mainstream CLT discusses element interactivity, but usually does not represent instructional material as nodes, weights, and relations.

No partitioning objective

Segmentation is treated as a design principle, but not as an optimization problem with local relief and global coherence loss.

Weak theory-measurement bridge

Multimedia learning work often invokes working memory and cognitive load without specifying how constructs map to outcomes.

Anchors: Sweller, 2010; Chen et al., 2023; Anmarkrud et al., 2019; Sawicka, 2008

Proposal

Treat instructional design as a constrained partitioning problem over structured information.

structured material

→

capacity-limited processing

→

partition choice

→

recall + transfer

The model asks: **how much should material be segmented before the cost of fragmentation exceeds the benefit of local relief?**

Core Tension

Segmentation is useful because it lowers local processing demand.

But segmentation is risky because it can break the relations needed for transfer.

Instructional partitioning

local relief

coherence loss

good design = tractable segments + preserved structure

Conceptual bridge: element interactivity + coherence-based comprehension

Theoretical Bridge

Cognitive load theory

Intrinsic and extraneous cost; element interactivity; prior knowledge.

$$I_i, E_i, r_j$$

Working-memory resource models

Capacity as graded, allocable, and costly.

$$C_i, \text{ optimization}$$

Coherence and comprehension

Understanding depends on preserved relational structure.

$$L(\Pi), \text{ transfer}$$

Formal cognition models

Capacity limits can be modeled as constrained optimization.

objective function

Anchors: Baddeley, 1992; Ma et al., 2014; Kintsch, 1994; Sawicka, 2008

Memory Partition Demo

Memory intuition: the same ten digits can be presented as one unsegmented string, as useful chunks such as 739 · 481 · 6205, or as ten isolated digits. Moderate partitioning reduces local burden; excessive partitioning removes useful binding structure.

Representing Instructional Structure

Instructional material is represented as a structured set of information elements. A partition divides the set into instructional segments; strong relations split across segments create connection loss.

$$A = \{a_1, \dots, a_k\}$$

q_j = element complexity

w_{uv} = coordination demand

r_j = residual novelty

$\Pi = \{A_1, \dots, A_p\}$ = partition

Minimal Assumptions

A1 · Decomposability

Instructional material can be represented as elements and relations.

A2 · Learner-relative cost

Prior knowledge reduces residual novelty and therefore effective burden.

A3 · Relation cost

w_{uv} means simultaneous coordination demand, not generic semantic similarity.

A4 · Common scale

C_i , E_i , and I_i are treated as comparable latent processing quantities.

A5 · Partition trade-off

Good segmentation improves local tractability without destroying too much global structure.

Processing Cost

$$I_i = \alpha \sum_{a_j \in A_i} r_j + \beta \sum_{a_j \in A_i} q_j r_j + \gamma \sum_{\substack{a_u, a_v \in A_i \\ u < v}} w_{uv} r_u r_v$$

effective quantity

element complexity

coordination demand

$$P_i = (C_i - E_i - I_i)_+$$

Goal-relevant processing is what remains after capacity is consumed by extraneous and intrinsic cost.

Optimization Objective

$$\Pi^* = \arg \max_{\Pi} \left[\sum_{i=1}^p P_i - \lambda L(\Pi) \right]$$

The objective rewards accumulated goal-relevant processing and penalizes connection loss. Moderate segmentation can be optimal when it lowers local cost without cutting too many strong relations.

λ is a theoretical fragmentation-sensitivity weight, not yet a fully estimated empirical parameter.

Hypotheses

H1

When unsegmented material exceeds feasible capacity, moderate segmentation should improve learning.

H2

Excessive segmentation should hurt transfer more than recall because transfer depends on preserved relations.

H3

Integrated representations should improve outcomes by reducing extraneous cost.

H4: Learners with more prior knowledge should tolerate larger or denser chunks because residual novelty is lower.

$$Y_{\text{recall}} = f_R \left(\sum_i P_i \right) \quad Y_{\text{transfer}} = f_T \left(\sum_i P_i, L(\Pi) \right)$$

Validation + Contribution

Validation path

1. Choose material decomposable into elements and relations.
2. Compare segmentation and representation conditions.
3. Measure prior knowledge, recall, transfer, and optional subjective load.
4. Test whether model summaries align with directional outcomes.

Narrow novelty claim

Existing models optimize memory fidelity or resource allocation.

This proposal optimizes the partitioning of structured instructional information under capacity limits.

The project turns a familiar verbal problem in instructional design into a precise, constrained, and testable formal account.